

MUHAMMAD HARIS

+92 347 8405339 | mharis6673@gmail.com | linkedin.com/in/haris6219/ | github.com/MH612188-DS

AI/ML ENGINEER | GENERATIVE AI | DEEP LEARNING | LLM/RAG | PYTORCH

PROFESSIONAL SUMMARY

AI/ML Engineer and Robotics & AI master's researcher at NUST with hands-on experience developing machine learning, generative AI, and LLM-powered systems. Strong foundation in Python, PyTorch, deep learning, model evaluation, experimentation, dataset construction, and model adaptation, combined with practical experience integrating ML and AI systems into production-oriented APIs and backend services. Research experience includes conditional generative modeling, cross-subject evaluation, personalization, and failure analysis for EMG-conditioned prosthetic grasp generation. Experienced with FastAPI, PostgreSQL, vector databases, Docker, AWS, CI/CD, and reproducible ML workflows.

TECHNICAL SKILLS

AI/ML: Python, PyTorch, TensorFlow/Keras, scikit-learn, NumPy, Pandas, Transformers, Deep Learning, Generative AI, Model Adaptation, Machine Learning

ML Engineering & Research: Data Preprocessing, Dataset Construction, Feature Engineering, Model Training, Model Evaluation, Experiment Design, Ablation Studies, Statistical Analysis, Cross-Subject Evaluation, Failure Analysis

Backend & Deployment: FastAPI, REST APIs, PostgreSQL, SQL, Docker, Docker Compose, Linux, Git/GitHub, GitHub Actions, CI/CD, AWS

Research: Generative Modeling, Computer Vision, Signal Processing, EMG, Model Fine-Tuning & Adaptation

EXPERIENCE

AI Software Developer | DevCrafterAI LLC | 2025 – Present

Developed AI-enabled applications and backend services involving ML inference, REST APIs, PostgreSQL, Docker, automated testing, and AI/ML integration.

- **EduBuddy:** Built an ML inference service that preprocesses user and behavioral features, executes trained-model predictions, and exposes personalized outputs through a REST API.
- **GovComm:** Contributed to a secure full-stack communication platform using Flutter, Node.js/Express, PostgreSQL, Socket.IO, MFA, Docker Compose, and automated API/unit testing.
- **Financial Risk Engine:** Worked with a modular C++ financial risk engine, implementing data ingestion, quantitative calculations, reusable components, and testable software architecture.
- I built and evaluated a real-time voice-agent prototype using STT → LLM/RAG → tool calling → TTS, measuring latency, transcription quality and response reliability.

RESEARCH & SELECTED AI / ML PROJECTS

Generative AI for Biomimetic Prosthetic Hand Grasping | NUST

- Developing conditional VAE and diffusion models for EMG-conditioned prosthetic grasp trajectory generation, with emphasis on personalization and cross-subject generalization.
- Built reproducible ML pipelines covering EMG preprocessing, dataset construction, model training, adaptation, cross-subject evaluation, and failure analysis.
- Conducted controlled experiments and ablation studies to evaluate model generalization, personalization, and intent-conditioned trajectory generation.

GandharaGen | Generative AI Image Synthesis Platform

- Built an end-to-end generative AI application combining historical-document retrieval, LLM-based context enrichment, and GPU-based image generation.
- Developed a RAG pipeline using SentenceTransformers embeddings and FAISS IndexFlatL2 for semantic top-k retrieval from historical text corpora.
- Integrated Gemini 1.5 Flash for context-grounded prompt generation and Hugging Face Diffusers with PyTorch/CUDA for local image generation.

CortexOS | AI Knowledge & RAG Platform

- Built a modular Python AI knowledge platform with multi-source ingestion, semantic chunking, embeddings, vector retrieval, PostgreSQL persistence, and provider-agnostic LLM integration.
- Implemented hybrid retrieval combining semantic vector search and lexical relevance, with adapters for FAISS, Qdrant, and PostgreSQL/pgvector.
- Integrated OpenAI, Claude, Gemini, and Ollama through provider-agnostic abstractions, enabling interchangeable model providers and consistent application interfaces.
- Containerized the platform with Docker and implemented CI/CD using GitHub Actions, deploying the FastAPI application to AWS EC2 with ECR, Systems Manager, IAM, and CloudWatch monitoring.

EDUCATION

MS Robotics & AI Engineering | NUST | 2024 - Present

BS Mechatronics Engineering | NUST | 2018 - 2022